

Initial Project Planning Template

Date	3 July 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Sprint Backlog

Sprint	Epic	User Story	Assigned To	Priority	Status
Sprint 1	Data Collection	As a data engineer, I can download the Kaggle credit card dataset so that the team has raw data to work with	sirisha	High	Done
Sprint 1	EDA	As a data analyst, I can explore application_record.csv and credit_record.csv to understand data distribution and quality	satya	High	Done
Sprint 2	Target Engineering	As an ML engineer, I can map STATUS codes to binary labels so that the model has a meaningful prediction target	hansraj	High	Done
Sprint 2	Feature Engineering	As an ML engineer, I can derive AGE_YEARS, YEARS_EMPLOYED, IS_UNEMPLOYED, and INCOME_PER_MEMBER features so that the model has richer inputs	prasanna	High	Done

Sprint	Epic	User Story	Assigned To	Priority	Status
Sprint 2	Preprocessing	As an ML engineer, I can apply LabelEncoder and StandardScaler and save them as PKL files so that the web app can preprocess user inputs consistently	sravya	High	Done
Sprint 3	Model Training	As a data scientist, I can train Logistic Regression, Decision Tree, Random Forest, and XGBoost classifiers so that we can compare model performance objectively	sirisha	High	Done
Sprint 3	Model Evaluation	As a data scientist, I can compare accuracy and ROC-AUC for all 4 models and select the best one so that the web app uses the most accurate predictor	satya	High	Done
Sprint 4	Flask Backend	As a backend developer, I can implement 4 Flask routes (/, /predict GET+POST, /history, /clear_history) so that users can interact with the model through a web browser	hansraj	High	Done
Sprint 4	Frontend Development	As a frontend developer, I can build a responsive Bootstrap 5 UI with a prediction form, result card, and history page so that the app is easy to use on any device	satya	High	Done
Sprint 4	Scenario Cards	As a frontend developer, I can add pre-fill scenario buttons (Low/Mid/High risk) so that demo users can quickly test different applicant profiles	sravya	Medium	Done
Sprint 5	Testing	As a QA engineer, I can test all form scenarios and edge cases so that the application delivers correct results across all input combinations	prasanna	High	Done
Sprint 5	Deployment	As a DevOps engineer, I can configure Render with Procfile and render.yaml so that the application is accessible at a public HTTPS URL	sirisha	High	Done

Sprint	Epic	User Story	Assigned To	Priority	Status
Sprint 5	Documentation	As a project lead, I can create all 8-phase documentation files and push them to GitHub so that the project is fully documented for submission	All	High	Done

Project Timeline

Phase	Duration	Deliverables
Sprint 1: Data & EDA	Week 1	Kaggle dataset, Jupyter EDA notebook
Sprint 2: Feature Engineering	Week 2	Preprocessed dataset, PKL encoders/scalers
Sprint 3: Model Training	Week 3	4 trained models, model.pkl, metadata.pkl
Sprint 4: Web App	Week 4	Flask app, all templates, static assets
Sprint 5: Testing & Deploy	Week 5	Render deployment, docs, GitHub push

Team Assignment Summary

Team Member	Role	Primary Responsibility
sirisha	Team Lead / ML Engineer	Target engineering, model training, frontend
prasanna	Backend Developer	Flask routes, prediction logic
satya	Data Engineer	Data collection, preprocessing, deployment
sravya	Data Analyst	EDA, testing, validation
hansraj	ML Engineer	Feature engineering, model evaluation

